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1 Concepts and Theorems

- $(a \pm b)^2 = a^2 \pm 2ab + b^2$
- $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$.
- (Binomial Theorem)

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}.$$

- $a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \dots + ab^{n-2} + b^{n-1})$.

Definition 1.1. If $b = aq$ for some $q \in \mathbf{Z}$, Then we say a divides b , denoted as $a \mid b$.

Theorem 1.2 (Divisibility). Let $a, b \in \mathbf{Z}$.

(a) $a \mid b, b \mid c \Rightarrow a \mid c$.

(b) $d \mid a, d \mid b \Rightarrow d \mid ax + by$ for all $x, y \in \mathbf{Z}$, In particular $d \mid a + b, d \mid a - b$.

(c) If any two terms in $a + b = c$ are divisible by d , then the 3rd term is also divisible by d .

Proof. We will prove (a) and leave the rest as exercises. Suppose $a \mid b$, and $b \mid c$, then $b = aq_1, c = bq_2$ for some $q_1, q_2 \in \mathbf{Z}$. Hence $c = aq_1q_2$, hence $a \mid c$ for $q_1q_2 \in \mathbf{Z}$. $\square$

Theorem 1.3 (Division Algorithm). For any $a \in \mathbf{Z}$, When a is divided by b , then $a = qb + r$ where $0 \leq r < |b|$. This representation is unique.

Definition 1.4. Suppose $a, b \in \mathbf{Z}$ are non-zero. The greatest common divisor (gcd) is the greatest number d such that $d \mid a$ and $d \mid b$, denoted $\gcd(a, b) = d$ or $(a, b) = d$. The least common multiple (lcm) is the least number ℓ such that $a \mid \ell$ and $b \mid \ell$, denoted $\text{lcm}(a, b) = \ell$ or $[a, b] = \ell$.

Theorem 1.5. For all $a, b \in \mathbf{Z}$, there exists $x, y \in \mathbf{Z}$ such that

$$ax + by = d = (a, b).$$

In particular, if $(a, b) = 1$, then there exists $x, y \in \mathbf{Z}$ such that

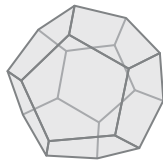
$$ax + by = 1.$$

Theorem 1.6. For prime p , $p \nmid \binom{p}{k}$, $0 < k < p$.

Theorem 1.7 (Fermat's little Theorem). Let p be a prime and $p \nmid a$, then:

$$a^{p-1} \equiv 1 \pmod{p}.$$

Corollary 1.8. Let p be prime, then for all integers a, b , $(a + b)^p \equiv a^p + b^p \pmod{p}$



Definition 1.9. $a \equiv b \pmod{m}$ if and only if $a - b$ is divisible by m ($m \mid a - b$)

Theorem 1.10 (同余基本). For $a, b, c, d, m \in \mathbf{Z}$, we have:

1. $a \equiv b \pmod{m}, b \equiv c \pmod{m} \Rightarrow a \equiv c \pmod{m}$
2. $a \equiv b \pmod{m}, c \equiv d \pmod{m} \Rightarrow a \pm c \equiv b \pm d \pmod{m}$.
3. $a \equiv b \pmod{m}, c \equiv d \pmod{m} \Rightarrow ac \equiv bd \pmod{m}$.
4. $ab \equiv ac \pmod{m}, (a, m) = 1 \Rightarrow b \equiv c \pmod{m}$.

Notation 1. $N = \overline{a_n a_{n-1} \dots a_0}$, $a_i \in \{0, 1, 2, \dots, 9\}$ represents an $n + 1$ digit number.

Example 1.1. $N = \overline{a_n a_{n-1} \dots a_0} = a_n \times 10^n + \dots + 10 \times a_1 + a_0$.

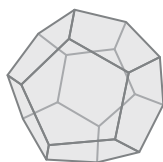
Theorem 1.11. Let $N = \overline{a_n a_{n-1} \dots a_0}$.

1. $N \equiv 0 \pmod{3}$ iff $a_n + a_{n-1} + \dots + a_0 \equiv 0 \pmod{3}$.
2. $N \equiv 0 \pmod{9}$ iff $a_n + a_{n-1} + \dots + a_0 \equiv 0 \pmod{9}$.
3. $N \equiv 0 \pmod{11}$ iff $a_0 - a_1 + a_2 - \dots + (-1)^n a_n \equiv 0 \pmod{11}$.
4. $N \equiv 0 \pmod{4}$ iff $4 \mid \overline{a_1 a_0}$.
5. $N \equiv 0 \pmod{8}$ iff $8 \mid \overline{a_2 a_1 a_0}$.



2 Examples and Problems

Problem 1. Find all $n \in \mathbb{Z}$ for which $n^2 + n + 1 \mid n^3 + 2n^2 - n - 3$.



Problem 2. Find all $a, b \in \mathbf{N}$ such that $a \mid b + 1$ and $b \mid a + 1$.



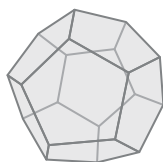
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Problem 3. Let $a, b \in \mathbb{N}$ such that $a \mid b^2 + 1$ and $b \mid a^2 + 1$. Prove that $ab \mid a^2 + b^2 + 1$.



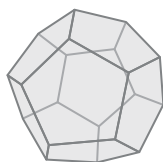
Problem 4. Prove that $8 \mid n^2 - 1$ for all odd integer n .



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Problem 5. For $n \in \mathbf{N}$ and $n > 1$. Find $(n^{30} - 1, n^{35} - 1)$.



Problem 6. Prove that $(n^3 + n^2 + 2n + 1, n^2 + n + 1) = 1$ for all $n \in \mathbf{Z}$.



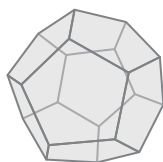
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Problem 7. Assume $a, b \in \mathbf{N}$ are relatively prime. Show that there are no $m, n \in \mathbf{N}$ such that $am + bn = ab$.



Problem 8. Prove that $n^3 \equiv n \pmod{6}$ for all $n \in \mathbf{Z}$.

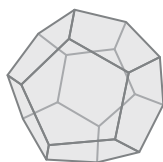


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Problem 9. Solve the following congruence equations:

- (a) $3n \equiv 4 \pmod{7}$.
- (b) $4n \equiv 1 \pmod{6}$.
- (c) $15n \equiv 5 \pmod{10}$.



Problem 10. Solve the following system of congruence equations:

(a)
$$\begin{cases} n \equiv 2 \pmod{3} \\ n \equiv 5 \pmod{7} \end{cases}$$

(b)
$$\begin{cases} 3n \equiv 4 \pmod{17} \\ 2n \equiv 3 \pmod{19} \end{cases}$$

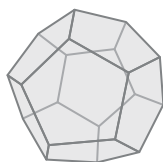
(c)
$$\begin{cases} 2n \equiv 2 \pmod{3} \\ 3n \equiv 5 \pmod{4} \\ 2n \equiv 3 \pmod{5} \end{cases}$$



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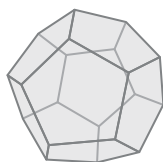
Problem 11. Prove that $2^{4n} - 1$ is divisible by 15 for all $n \in \mathbb{N}$.



Problem 12. Prove that $11^{n+2} + 12^{2n+1}$ is divisible by $133 = 7 \cdot 19$ for all $n \in \mathbf{N}$.



Problem 13. Let n be a positive integer composed of exactly 300 1's and various number of zeros. Prove that n is not a perfect square.



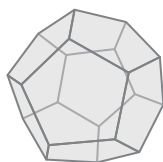
Problem 14. (IMO 1964 Q1)

- (a) Find all $n \in \mathbf{N}$ such that $2^n - 1$ is divisible by 7.
- (b) Prove that $2^n + 1$ is not divisible by 7 for all $n \in \mathbf{N}$.



Problem 15.

- (a) Prove that 7 cannot be written as a sum of 3 squares.
- (b) Prove that $n^2 \equiv 0, 1, 4 \pmod{8}$ for all $n \in \mathbf{N}$.
- (c) (IMO 1969 Long List) Prove that there are infinitely many $n \in \mathbf{N}$ that cannot be written as a sum of three squares.



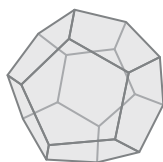
Problem 16. Given a, b and $a - b$ are positive integers not divisible by 3. Prove that $a^3 + b^3$ is divisible by 9.



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Problem 17. Given $5 \mid a + b$, prove that $25 \mid a^5 + b^5$.



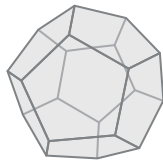
Problem 18. Let p be a prime and that $a \equiv b \pmod{p}$. Prove that $a^p \equiv b^p \pmod{p^2}$.



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Problem 19. Let p be a prime such that $p \equiv 2 \pmod{3}$. Prove that for all $n \in \mathbf{Z}$, there exists $a \in \mathbf{Z}$ such that $n \equiv a^3 \pmod{p}$.



Problem 20. Find the total number of odd numbers in the 100th row of Pascal's triangle.



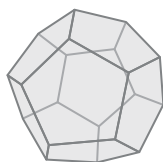
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Problem 21. Given $(1+x)^{38} = a_0 + a_1x + \cdots + a_{38}x^{38}$. Let $N_k = \#\{a_i \mid a_i \equiv k \pmod{3}\}$. Find the value of $N_1 - N_2$.



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Problem 22. Given $(1 + x + y)^{17} = a_0 + a_1x + a_2y + a_3x^2 + a_4xy + a_5y^2 + a_6x^3 + a_7x^2y + \cdots + a_ky^{17}$.

(a) Find k .

(b) Find the total number such that a_i is odd.



Problem 23. (IMO 2023 Q1) Determine all composite integers $n > 1$ that satisfy the following property: if $d_1, d_2, \dots, d_k$ are all the positive divisors of n with $1 = d_1 < d_2 < \dots < d_k = n$, then d_i divides $d_{i+1} + d_{i+2}$ for every $1 \leq i \leq k - 2$.